Catalogue of ODE specifications

Extracted from a doctoral thesis on stochastic branching and release processes

Neutral-terminology copy for automated review

September 2026

Companion to `markov_models_neutral.tex`. Every entry gives the thesis location, the equations, parameters, and solutions or thresholds stated with them.

Contents

Scope and conventions

This document lists every deterministic system of ordinary differential equations written down in the source thesis, in four classes:

1. **Standalone population models** specified as deterministic systems (D1–D8);
2. **Flow components of hybrid ODE–CTMC models** — the deterministic part of a piecewise-deterministic pair whose jump part is catalogued in the companion Markov document (D9–D12);
3. **Moment and backward ODE systems** derived from the Markov models and written down there as equations in their own right (D13–D16);
4. **Continuum approximations** replacing a recursion by an ODE (D17).

One boundary case is included with a flag: the renewal system D8 is integro-differential rather than purely ODE, but it is the thesis's main deterministic population model and reduces to ODE form in its exponential-phase projection. Partial derivative equations (generating function PDEs, transport equations) are not ODEs and are omitted. Cross-references D n are internal; M n refer to the companion catalogue. Neutral vocabulary throughout: *compartment* (founded, active, or cleared), *free particles*, *target sites*, *founding rate*.

Chapter 1

Catalogue at a glance

	System	Variables	Where
Standalone deterministic models			
D1	Standard population model, three variables	$T, \mathcal{I}, \mathcal{V}$	§1.5
D2	Standard population model, two variables	$\mathcal{I}, \mathcal{V}$	§1.5; §6.2.1
D3	Mean-field seasonal two-population system	x, y	§2.2.3.1
D4	Logistic environment ODE	N	§2.2.3.2; §9.2
D5	Gated-release flow system (hybrid flow part)	A, R + gate G	§2.2.4.1
D6	Coupling taxonomy: three vector-field forms	$\mathbf{y}$ $(, M_t)$	§2.2.4
D7	Partial-release mean-field system	$\mathcal{I}, Q, \mathcal{V}$	§6.7.2
D8	Renewal system (integro-differential; flagged)	$\mathcal{I}, \mathcal{V}$ + kernels	§6.4
Flow components of hybrid models			
D9	RRRR potential flow	V	§9.4.2
D10	Multiplicative-dissipation potential flow	V_i	§9.5.4
D11	Vitality flow (birth–conversion variant)	V	§9.A
D12	Public-good concentration flow	P	§9.B
Moment and backward ODE systems			
D13	Birth–death moment ODEs (+ backward Riccati)	moments; u	§2.2.3; §2.3.2
D14	Birth–death–catastrophe moment system	I, J, K, V	§4.4; §5.2.4
D15	Master-equation (forward) infinite systems	$p_j(t)$	§2.3.2.1; §5.3.1
D16	Two-type backward Riccati system	S, G	§7.2.3–7.2.4
Continuum approximation			
D17	Continuum ODE for the survival recursion	S	§3.7

Chapter 2

Standalone deterministic models

2.1 D1: Standard population model, three variables

Specified in §1.5.

Variables. Unoccupied target sites T, active compartments $\mathcal{I}$, free particles $\mathcal{V}$.

Equations.

$$\frac{\mathrm{d}T}{\mathrm{d}t} = s - \gamma T\mathcal{V}, \qquad \frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} = \gamma T\mathcal{V} - d_I\,\mathcal{I}, \qquad \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} = p\,\mathcal{I} - c\,\mathcal{V}, \tag{2.1}$$

with site influx s, founding rate γ per site–particle pair, release rate p per active compartment, removal rate d_I, particle clearance c.

Role. The thesis carries the target-constant reduction D2; D1 is its full form, shown for context in the opening chapter.

2.2 D2: Standard population model, two variables (target-constant)

Specified in §1.5 (reduction); §6.2.1 as the standard comparator.

Equations.

$$\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} = \gamma T\,\mathcal{V} - d_{\mathcal{I}}\,\mathcal{I}, \qquad \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} = p\,\mathcal{I} - c\,\mathcal{V}, \qquad R_0 = \frac{\gamma T\,p}{c\,d_{\mathcal{I}}}, \tag{2.2}$$

with target sites T held fixed (large-site regime: site depletion is not a driver of the dynamics).

Status and anchors. Deterministic; the thesis shows it is the exact one-compartment limit of the budding comparator (M15) and a projection of the renewal system D8, valid in exponential phase. Every active compartment, whatever its age or contents, releases at rate p and is removed at rate $d_{\mathcal{I}}$ — the two constants that D8 replaces by kernels.

2.3 D3: Mean-field seasonal two-population system

Specified in §2.2.3.1, equation following the seasonal generator.

Equations. Deterministic comparison system for the two-count process M7:

$$\dot{x} = b\,\mathcal{S}^{+}(t)\,x - exy, \qquad \dot{y} = exy - dy, \qquad \mathcal{S}^{+}(t) = \max\{0,\, 1 + A\sin(\omega t)\}. \tag{2.3}$$

Anchors. For $A > 1$ each period $2\pi/\omega$ contains a closed season of length $(2/\omega)\arccos(1/A)$. On the axis $y = 0$ the exact mean of M7 is $x_0 \exp\left(b\int_0^t \mathcal{S}^{+}(s)\mathrm{d}s\right)$; in the interior the stochastic first moments involve $\mathbb{E}(xy)$ and do not close, which is why the deterministic system is only a comparison. Figure parameters: $A = 2$, $\omega = 2\pi$, $b = 2$, $e = 0.005$, $d = 2$, $(x_0, y_0) = (300, 100)$.

2.4 D4: Logistic environment ODE

Specified in §2.2.3.2; §9.2 (Chapter 9 notation).

Equation.

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \varrho N\left(1 - \frac{N}{K}\right), \qquad N(0) = N_0, \qquad 0 < N_0 < K, \tag{2.4}$$

with carrying capacity K and growth rate ϱ (Chapter 2 uses r, C).

Solution.

$$N(t) = \frac{K\,\mathrm{e}^{\varrho t}}{\xi + \mathrm{e}^{\varrho t}}, \qquad \xi = \frac{K}{N_0} - 1. \tag{2.5}$$

Role. Prescribed environment driving the type-count process M8: the per-type origination rate is $\beta(t) = \sigma(K - N(t))$; the coupling is one-way. The type-count mean obeys $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(S_t) = \beta(t)\mathbb{E}(S_t)$ (§9.2.5), giving $\mathbb{E}(S_t) = \mathrm{e}^{\mathcal{B}(t)}$ with $\mathcal{B}(t) = \int_0^t \beta$ and the limit $\mathbb{E}(S_\infty) = (K/N_0)^{\sigma K/\varrho}$.

2.5 D5: Gated-release flow system

Specified in §2.2.4.1.

Variables. Accumulated amount A_t, transferred amount R_t, gate $G_t \in \{0, 1\}$ (jump part: M9).

Flow (between gate jumps).

$$\frac{\mathrm{d}A_t}{\mathrm{d}t} = \rho - G_t\,cA_t, \qquad \frac{\mathrm{d}R_t}{\mathrm{d}t} = G_t\,cA_t, \tag{2.6}$$

external input $\rho > 0$, transfer rate $c > 0$.

Anchors. Pathwise balance $A_t + R_t = A_0 + R_0 + \rho t$; both components stay non-negative and R_t is non-decreasing. The first moment $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(A_t) = \rho - c\mathbb{E}(G_t A_t)$ is not closed. With the gate held fixed the flows solve exactly: closed, A grows linearly; open, $A \to \rho/c$. Illustration parameters: $\rho = 2$, $c = 4$, $\eta = 0.015$, $\gamma = 1.45$, $(A_0, R_0, G_0) = (5, 0, 0)$.

2.6 D6: Coupling taxonomy: three vector-field forms

Specified in §2.2.4.

Specification. With continuous state $\mathbf{y}(t) \in \mathbb{R}^d$ and discrete mode M_t, the thesis writes the three couplings as

$$\frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t} = F(\mathbf{y}(t), M_t), \qquad M_t \text{ autonomous} \qquad \text{(case i; M10)}, \tag{2.7}$$

$$\frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t} = F(\mathbf{y}(t)), \qquad q_{ij}(t) = q_{ij}(\mathbf{y}(t)) \qquad \text{(case ii; M8)}, \tag{2.8}$$

$$\frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t} = F(\mathbf{y}(t), M_t), \qquad q_{ij} = q_{ij}(\mathbf{y}(t)) \qquad \text{(case iii; M9)}. \tag{2.9}$$

In all three cases the pair is a piecewise-deterministic Markov process; in case (iii) generally only the pair is Markov. Case (ii) can be solved for $\mathbf{y}$ first, leaving a time-inhomogeneous chain.

2.7 D7: Partial-release mean-field system

Specified in §6.7.2.

Variables. Active compartments $\mathcal{I}$, total intracompartmental stock Q across productive compartments, free particles $\mathcal{V}$.

Equations. Release events on the load-dependent clock δX with each unit released independently with probability φ:

$$\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} = \gamma T\,\mathcal{V} - d_{\mathcal{I}}\mathcal{I}, \qquad \frac{\mathrm{d}Q}{\mathrm{d}t} = \nu\mathcal{I} - \delta\varphi\,\frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}}Q, \qquad \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} = \delta\varphi\,\frac{Q^2}{\mathcal{I}} - c\,\mathcal{V}, \tag{2.10}$$

with immigration ν into newly founded compartments.

Anchors. With a load-independent clock $\lambda(X) = \delta$ instead, mean export is $\delta\varphi Q$ and the system collapses to stock-and-export dynamics with $\varepsilon_{\mathrm{eff}} = \delta\varphi$: partial release is invisible in the mean field exactly when the clock is load-independent, and visible in single-compartment release-count distributions always. Mapping used in the thesis: blocking production lowers λ; enhancing degradation raises μ; blocking founding lowers γT; blocking secretion reduces φ.

2.8 D8: Renewal system (integro-differential; flagged)

Specified in Kernels §6.3; system §6.4.1; skeleton §6.4.3.

Specification. Not an ODE system — a pair of Volterra-type renewal equations — included because it is the thesis's principal deterministic population model and reduces to ODE/algebraic form under exponential growth. With incidence $i(t) = \gamma T\mathcal{V}(t)$ and $\mathcal{I}_0$ initially active compartments of age zero:

$$\mathcal{I}(t) = \mathcal{I}_0\, I_{\mathrm{fix}}(t) + \int_0^t i(t-\alpha)\, I_{\mathrm{fix}}(\alpha)\,\mathrm{d}\alpha = \mathcal{I}_0\, I_{\mathrm{fix}} + (i * I_{\mathrm{fix}})(t), \tag{2.11}$$

$$\frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t}(t) = \mathcal{I}_0\, g(t) + \int_0^t i(t-\alpha)\, g(\alpha)\,\mathrm{d}\alpha - c\,\mathcal{V}(t) = \mathcal{I}_0\, g + (i * g)(t) - c\,\mathcal{V}(t). \tag{2.12}$$

Kernels from M6: $S(\alpha) = I_{\mathrm{fix}}(\alpha)$ and $g(\alpha) = \delta K(\alpha)$.

Skeleton and anchors. Any kernel pair (S, g) induces $\mathcal{I}(t) = \mathcal{I}_0 S(t) + (i * S)(t)$ and $\frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} = \mathcal{I}_0 g + (i * g) - c\mathcal{V}$. Under exponential incidence $i(t) = i_0 \mathrm{e}^{rt}$ the system reduces to algebraic effective parameters $p_{\mathrm{eff}}(r) = \widetilde{g}(r)/\widetilde{S}(r)$ and, for linear incidence, $R_0 = (\gamma T/c)\int_0^\infty g$; exponential kernels reproduce D2 exactly. First-moment exactness (Proposition, §6.4.2) holds because all rates are linear.

Chapter 3

Flow components of hybrid ODE–CTMC models

Each entry below is the deterministic flow of a piecewise-deterministic pair; the jump process is catalogued as M27, M28, M29 (variant), and M30 respectively.

3.1 D9: RRRR potential flow

Specified in §9.4.2.

Equation.

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \phi - c\,X_t, \tag{3.1}$$

potential $V(t)$, inflow ϕ, consumption c per agency, with X_t the jump process of M27.

Anchors. V cannot be exhausted: as $V \downarrow 0$ with $X_t \geq 1$ the catastrophe hazard behaves like $\delta/\,(c(t^* - t))$, which integrates to infinity, so the catastrophe arrives first and V stays positive for all time. Illustration parameters: $\phi = 4$, $c = 0.8$.

3.2 D10: Multiplicative-dissipation potential flow

Specified in §9.5.4.

Equation. For each extant type i of the automaton M28:

$$\frac{\mathrm{d}V_i}{\mathrm{d}t} = \varphi N_i - c\,X_i, \qquad \text{floored at } 0, \tag{3.2}$$

with occupied cells N_i (each supplying potential at rate φ) and extant sub-types X_i (each costing c).

Anchors (three readings stated in the source). (i) V_i does not appear on the right: a pure accumulator, growing while $X_i < \varphi N_i/c$ and falling beyond. (ii) Since $X_i \leq N_i$, the most negative drift is $N_i(\varphi - c)$: potential can fall at all only if $c > \varphi$, the condition under which fragmentation is reachable. (iii) A single-sub-type type needs at least c/φ cells to break even. The flow differs from D9 in exactly two respects: inflow charged per cell (endogenous) and outflow charged per type rather than per individual.

3.3 D11: Vitality flow (birth–conversion variant)

Specified in Appendix 9.A.

Equation. Carried alongside the birth–conversion pair (M29 variant):

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \phi - c\,X_t, \tag{3.3}$$

and the middle jump channel becomes $\mathbb{P}(\Delta X_t = -1,\ \Delta Y_t = 1) = (\delta/V(t) + \chi Y_t)\,X_t\,\mathrm{d}t$.

Anchors. A population that grows large drives V down and raises its own loss rate — the amplification the unvarianted model lacks. Same functional form as D9; different jump process. Not simulated.

3.4 D12: Public-good concentration flow

Specified in Appendix 9.B.

Equation.

$$\frac{\mathrm{d}P}{\mathrm{d}t} = g\,X_t - c, \tag{3.4}$$

public-good concentration P, production g per unit, constant clearance c, with the jump process of M30 (birth rate $\lambda(t) = \lambda_0 + \alpha P$).

Anchors. While $X_t < c/g$ the good is cleared faster than it is made and P falls, so a small founding cohort may never start the feedback loop; the regime of interest is $\mu > \lambda_0$ (subcritical alone, supercritical once enough good has accumulated). Illustration parameters: $(\lambda_0, \alpha, g, c) = (0.6, 0.25, 0.15, 0.45)$, so P falls whenever $X_t < c/g = 3$.

Chapter 4

Moment and backward ODE systems

4.1 D13: Birth–death moment ODEs, and the backward Riccati equation

Specified in Mean and second moment: §2.3.2.2–2.3.2.3; time-inhomogeneous mean and backward Riccati: §2.2.3.

Moment equations. For the linear birth–death process (M5), from the master equation D15:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(X_t) = (\lambda - \mu)\mathbb{E}(X_t)\,, \qquad \mathbb{E}(X_t) = N\mathrm{e}^{(\lambda-\mu)t}, \tag{4.1}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\left(X_t^2\right) = 2(\lambda{-}\mu)\mathbb{E}\left(X_t^2\right){+}(\lambda{+}\mu)\mathbb{E}(X_t)\,, \qquad \mathrm{Var}(X_t) = N\,\frac{\lambda+\mu}{\lambda-\mu}\,\mathrm{e}^{(\lambda-\mu)t}\left(\mathrm{e}^{(\lambda-\mu)t}-1\right)\ (\lambda \neq \mu), \tag{4.2}$$

with $\mathrm{Var}(X_t) = 2N\lambda t$ at criticality $\lambda = \mu$. The ϱth moment equation is driven by the lower moments, so the hierarchy is solved recursively; for nonlinear rates it does not close. The continuous-time mean also appears in §1.3 as the definition of the three regimes (subcritical, critical, supercritical by the sign of $\lambda - \mu$).

Time-inhomogeneous mean. For rates $\lambda(t)$, $\mu(t)$: $\mathbb{E}(X_t \mid X_0 = n_0) = n_0 \exp\left(\int_0^t (\lambda(r) - \mu(r))\,\mathrm{d}r\right)$, still closed by linearity.

Backward Riccati equation for extinction. With $u(s,T) = \mathbb{P}(X_T = 0 \mid X_s = 1)$:

$$-\frac{\partial u}{\partial s} = \mu(s) - (\lambda(s) + \mu(s))u + \lambda(s)u^2, \qquad u(T,T) = 0. \tag{4.3}$$

The forward-endpoint probability $u_{\mathrm{f}}(t) = B(t)/(1{+}B(t))$ with $B(t) = \int_0^t \mu(s)\exp\left(\int_0^s (\mu(r) - \lambda(r))\mathrm{d}r\right)\mathrm{d}s$ does *not* satisfy this equation with s replaced by t; the two time endpoints play different roles.

4.2 D14: Birth–death–catastrophe moment system

Specified in Riccati equation and mean: §4.3, §5.2.4; moment hierarchy and release balance: §4.4.

Core Riccati equation. The no-catastrophe probability $I(t)$ of M6 obeys the closed scalar ODE

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \lambda(I-a)(I-b) = \mu - (\lambda+\mu+\delta)I + \lambda I^2, \qquad I(0) = 1, \tag{4.4}$$

with roots $b < 1 < a$ from $\eta = (\lambda+\mu+\delta)/(2\lambda)$, $a, b = \eta \pm \sqrt{\eta^2 - \mu/\lambda}$. $D(t)$ satisfies the same equation with $D(0) = 0$.

Moment balance equations. With $J(t) = \mathbb{E}(X_t)$, $K(t) = \mathbb{E}(X_t^2)$, and $V(t) = \mathbb{E}(W_t)$:

$$\frac{\mathrm{d}I}{\mathrm{d}t} = -\delta J, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{P}(X_t = H) = \delta J, \tag{4.5}$$

$$\frac{\mathrm{d}J}{\mathrm{d}t} = (\lambda - \mu)J - \delta K, \qquad \frac{\mathrm{d}V}{\mathrm{d}t} = \delta K, \tag{4.6}$$

and the genuinely open continuation $\frac{\mathrm{d}K}{\mathrm{d}t} = 2(\lambda-\mu)K + (\lambda+\mu)J - \delta\mathbb{E}(X_t^3)$. The quadratic term $-\delta K$ (and the cubic beyond) is the signature of load-proportional catastrophe: a large load is both more costly when removed and more likely to be removed.

Closure without truncation. The hierarchy is not truncated: from $J = -I'/\delta$ and the Riccati equation, $\frac{\mathrm{d}J}{\mathrm{d}t} = -(\lambda+\mu+\delta)J + 2\lambda IJ$, which eliminates K (and, by repeated differentiation, all higher moments) in favour of I. Closed forms at $w = \mathrm{e}^{\theta t}$: $J = \frac{(a-b)^2 w}{(B+Aw)^2}$, $K = \left(1 + \frac{2\lambda}{\delta}(1-I)\right)J$, $V = (1-I)\left(1 + \frac{\lambda}{\delta}(1-I)\right)$, and the second release moment $\mathbb{E}(W_t^2) = \frac{2(\lambda-\mu)}{\delta}V - \frac{\lambda+\mu}{\delta}I - K + \frac{\lambda+\mu}{\delta} + 1$. The balance identity $\frac{\mathrm{d}J}{\mathrm{d}t} = (\lambda-\mu)J - \frac{\mathrm{d}V}{\mathrm{d}t}$ also gives $\frac{\mathrm{d}V}{\mathrm{d}t} = -\frac{\lambda-\mu}{\delta}\frac{\mathrm{d}I}{\mathrm{d}t} - \frac{\mathrm{d}J}{\mathrm{d}t}$.

4.3 D15: Master-equation (forward Kolmogorov) infinite systems

Specified in General form and birth–death case: §2.3.2.1; BDC case, both rupture conventions: §5.3.1; absorption-model forward equations: §2.B.

General form. For a CTMC with generator Q and state probabilities $p_j(t) = \mathbb{P}(X_t = j)$:

$$\frac{\mathrm{d}p_j(t)}{\mathrm{d}t} = \sum_{k\neq j} q_{kj}\, p_k(t) - q_j\, p_j(t), \tag{4.7}$$

equivalently $\frac{\mathrm{d}P}{\mathrm{d}t} = PQ$; the backward system $\frac{\mathrm{d}P}{\mathrm{d}t} = QP$ carries the same information and suits functionals of the future.

Birth–death case (M5).

$$\frac{\mathrm{d}p_n(t)}{\mathrm{d}t} = \lambda(n-1)p_{n-1}(t) + \mu(n+1)p_{n+1}(t) - (\lambda+\mu)n\, p_n(t), \qquad n \geq 0, \tag{4.8}$$

an infinite coupled system; multiplying by z^n and summing collapses it to the generating-function PDE.

Birth–death–catastrophe case (M6), killing convention.

$$\frac{\mathrm{d}p_i}{\mathrm{d}t} = \lambda(i-1)p_{i-1} + \mu(i+1)p_{i+1} - (\lambda+\mu+\delta)ip_i \quad (i > 0), \qquad \frac{\mathrm{d}p_0}{\mathrm{d}t} = \mu p_1 + \delta\sum_{j\geq 1} jp_j = \mu p_1 + \delta J(t). \tag{4.9}$$

Catastrophe convention (thesis default). Rupture leaves the count altogether; the first equation then holds for all $i \geq 0$ with $p_{-1} = 0$, and

$$\frac{\mathrm{d}p_0}{\mathrm{d}t} = \mu p_1 : \tag{4.10}$$

internal extinction by death still feeds state 0, catastrophe does not. The generating function is defective, $G(1,t) = I(t) \leq 1$, under catastrophe and $G(1,t) = 1$ under killing.

4.4 D16: Two-type backward Riccati system

Specified in §7.2.3 (first-event derivation), §7.2.4 (dimensionless form).

Equations. For the no-catastrophe probabilities of M22, $S(t) = \mathbb{P}_{(1,0)}\{\tau_c > t\}$ and $G(t) = \mathbb{P}_{(0,1)}\{\tau_c > t\}$:

$$\frac{\mathrm{d}S}{\mathrm{d}t} = \lambda_1 S^2 - (\lambda_1 + \mu_1 + \nu + \delta_1)S + \mu_1 + \nu G, \qquad S(0) = 1, \qquad S'(0) = -\delta_1, \tag{4.11}$$

$$\frac{\mathrm{d}G}{\mathrm{d}t} = \lambda_2 G^2 - (\lambda_2 + \mu_2 + \delta_2)G + \mu_2, \qquad G(0) = 1, \qquad G'(0) = -\delta_2. \tag{4.12}$$

The system is triangular: G is autonomous, and its solution feeds the S equation, which is then reduced by a Riccati linearisation to the Gauss hypergeometric equation.

Dimensionless form. Time in units of λ_1 (assumed positive); scaled rates $\hat{\mu}_i = \mu_i/\lambda_1$, $\hat{\nu} = \nu/\lambda_1$, $\hat{\delta}_i = \delta_i/\lambda_1$, $\hat{\lambda}_2 = \lambda_2/\lambda_1$:

$$\widehat{S}' = \widehat{S}^2 - \hat{\kappa}_1 \widehat{S} + \hat{\mu}_1 + \hat{\nu}\widehat{G}, \qquad \hat{\kappa}_1 = 1 + \hat{\mu}_1 + \hat{\nu} + \hat{\delta}_1, \tag{4.13}$$

$$\widehat{G}' = \hat{\lambda}_2 \widehat{G}^2 - (\hat{\lambda}_2 + \hat{\mu}_2 + \hat{\delta}_2)\widehat{G} + \hat{\mu}_2. \tag{4.14}$$

The main formula assumes $\hat{\lambda}_2 > 0$, $\hat{\delta}_2 > 0$, $\hat{\nu} > 0$; the boundaries ($\lambda_1 = 0$, $\hat{\lambda}_2 = 0$, $\hat{\delta}_2 = 0$, $\hat{\nu} = 0$) are treated separately, and the zero-catastrophe case recovers the two-type branching equations of Antal and Krapivsky.

Chapter 5

Continuum approximation

5.1 D17: Continuum ODE for the survival recursion

Specified in §3.7 (paragraph "Continuum ODE").

Derivation setting. The binary Galton–Watson survival recursion $S_{n+1} = 2pS_n - pS_n^2$ (M2) with $p = \frac{1}{2} - \varepsilon/2$ (subcritical, $\varepsilon = 1 - 2p > 0$ small). Writing $S_{n+1} - S_n = -\varepsilon S_n - \frac{1}{2}S_n^2 + O(\varepsilon S_n^2)$ and treating generation number as continuous for large n:

$$\frac{\mathrm{d}S}{\mathrm{d}n} = -\varepsilon S - \tfrac{1}{2}S^2. \tag{5.1}$$

Solution. With $u = 1/S$: $u' = \varepsilon u + \frac{1}{2}$, so

$$S(n) = \frac{1}{C\,\mathrm{e}^{\varepsilon n} - \frac{1}{2\varepsilon}}, \qquad S(n) \simeq \frac{2\varepsilon}{\mathrm{e}^{\varepsilon n} - 1} \tag{5.2}$$

after matching to the critical decay $S_n \sim 2/n$, which fixes $C \sim 1/(2\varepsilon)$.

Anchors. Matching $S_n \sim A(p)(2p)^n \sim A\mathrm{e}^{-\varepsilon n}$ gives the near-critical amplitude $A(p) \sim 2\varepsilon = 2(1-2p)$ as $p \to \frac{1}{2}^-$, and $1/A(p) \sim 1/\left(2(1-2p)\right)$; the crossover generation $n \sim 1/\varepsilon$ separates the critical regime ($S \propto 1/n$) from the eventual geometric decay.